3 *The propagation speed of waves in different media*

3.1 The speed of a transverse wave in a stretched elastic string

Our goal here is to express the speed of a wave in a string in terms of parameters pertaining to the string, such as the tension, or the density of the string. Applying Newton's 2nd law to a tiny segment of string will be part of the process, so we begin by finding the force in the transverse direction at points along the string. Figure 3.1 shows a snapshot of a transverse harmonic wave in a stretched string. The blue arrows represent the transverse forces at different points along the string.

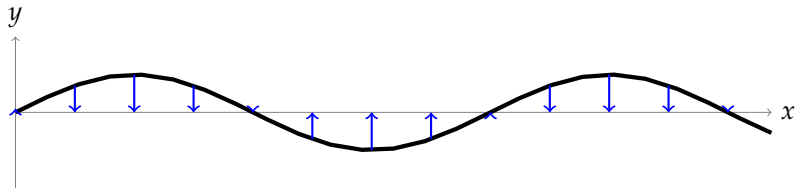


Figure 3.1: Transverse forces at different points along a vibrating string.

At each point along the string, the transverse force is different. This is because at each point, a small segment of the string is being pulled in different directions by the rest of the string to the right, and to the left.

Figure 4.1 shows a tiny segment at a certain position x, with the forces at each end of the segment. The net force on the segment is the sum of the two end forces.

Now in practice the angles are small, so to a good approximation (see "Approximating a function" in Appendix B)

$$\sin\theta \approx \tan\theta = \frac{\partial y}{\partial x} \tag{3.1}$$

Therefore the transverse force at x is

$$F_y(x) = -T\left(\frac{\partial y}{\partial x}\right)_x \tag{3.2}$$

and at $x + \delta x$ it is

$$F_y(x+\delta x) = +T\left(\frac{\partial y}{\partial x}\right)_{x+\delta x} \tag{3.3}$$

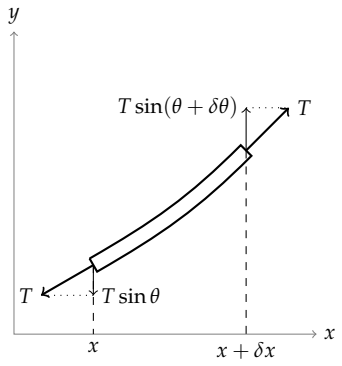


Figure 3.2: Segment of a stretched elastic string with a transverse wave moving through it. The slope of the string at x is θ, and the slope at $x + \delta x$ is $\theta + \delta\theta$.

The net transverse force on the segment is

$$T\left(\frac{\partial y}{\partial x}\right)_{x+\delta x} - T\left(\frac{\partial y}{\partial x}\right)_{x} \approx T\left(\frac{\partial^2 y}{\partial x^2}\right)\delta x \tag{3.4}$$

according to the result of equation (B.5) in Appendix B.

The net longitudinal force on the segment, on the other hand, is

$$T\cos(\theta + \delta\theta) - T\cos(\theta) \tag{3.5}$$

and in the small-angle approximation both cosines are approximately equal to 1, so the net longitudinal force is zero to a good approximation.

So we are left only with a net transverse force on the segment. This must be equal to the mass of the segment times its acceleration, according to Newton's 2nd law. If the mass per unit length of the string is μ, the mass of the segment is $\mu\delta x$; and its acceleration is $\partial^2 y/\partial t^2$. Putting everything together, Newton's 2nd law gives

$$\mu\delta x\frac{\partial^2 y}{\partial t^2} = T\left(\frac{\partial^2 y}{\partial x^2}\right)\delta x \tag{3.6}$$

or

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu}\left(\frac{\partial^2 y}{\partial x^2}\right) \tag{3.7}$$

If we compare this with the wave equation (2.14), we see that the propagation speed of the wave is

$$v = \sqrt{\frac{T}{\mu}} \tag{3.8}$$

Example 3.1: *A guitar string has a mass of* 2.00 g *and is* 85.0 cm *long. If the tension in the string is* 70.0 N*, what is the speed of a transverse wave in the string?*

The mass per unit length is $2.35 \times 10^3\,\mathrm{kg\,m^{-1}}$, which gives a speed of $173\,\mathrm{m\,s^{-1}}$.

3.2 *The speed of sound in a fluid*

An **acoustic wave**, or **sound wave** is a longitudinal compression wave. Again, for the moment we consider only one-dimensional waves, so imagine a long, narrow column of fluid aligned with an x axis.

Consider a layer in the column of fluid between x and $x + \delta x$ when the layer is unperturbed, as shown in the top diagram of Figure 3.3. If

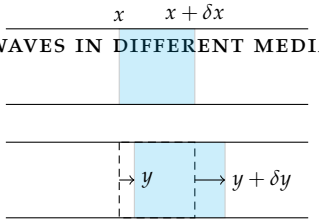


Figure 3.3: A layer in a column of fluid lies between x and $x + \delta x$ in equilibrium (top). As a longitudinal wave passes through the medium, the left boundary is displaced by y, and the right boundary by $y + \delta y$ (bottom).

the cross-sectional area of the column is A, the unperturbed volume is $A\delta x$.

As the wave passes through it, the left boundary is displaced by an amount y, and the right boundary by an amount $y + \delta y$, as indicated in the bottom diagram of the figure. Therefore the volume changes by an amount $A\delta y$.

Now for every fluid medium, there is a relation between pressure applied and the resulting volume contraction. It is characterized by the **bulk modulus** B of the medium, which is defined by

$$\delta p = -B\frac{\delta V}{V} \tag{3.9}$$

where δp is extra pressure applied, and $\delta V/V$ is the resulting fractional volume change. The minus sign is there because a positive pressure increment results in a negative volume change, as the fluid contracts. Thus, B is positive.

In this case, the fractional volume change is $A\delta y/A\delta x$, and therefore

$$\delta p = -B\frac{\delta y}{\delta x} \tag{3.10}$$

Now the net force on the layer is equal to the net pressure difference across it, multiplied by the cross-sectional area. If the pressure varies with x as shown in Figure 3.4, the net force on the layer is $-A\delta p$, and since $V = A\delta x$, the force is $-V\delta p/\delta x$. We will find it convenient to work with the force per unit volume, which is simply $\delta p/\delta x$.

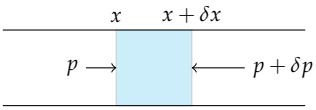


Figure 3.4: The pressure at the boundaries of the layer.

Substituting δp from equation (3.10), we find that the force per unit volume is given by

$$B\frac{1}{\delta x}\left(\frac{\delta y}{\delta x}\right) \tag{3.11}$$

If we make the increments infinitesimally small, the force per unit volume becomes

$$B\frac{\partial^2 y}{\partial x^2} \tag{3.12}$$

Newton's 2nd law says that the force is equal to the mass times the acceleration of the layer, and therefore the force per unit volume is equal to the *density* times the acceleration. The acceleration is $\partial^2 y/\partial t^2$, so we have

$$B\frac{\partial^2 y}{\partial x^2} = \rho\frac{\partial^2 y}{\partial t^2} \tag{3.13}$$

This is the wave equation! Comparing with equation (2.14) we find that the propagation speed is

$$v = \sqrt{\frac{B}{\rho}} \tag{3.14}$$

Example 3.2: *The bulk modulus of water is* $B = 2.2 \times 10^9$ Pa. *What pressure must be applied to decrease the volume of a sample of water by 1%? What is this in atmospheres if 1 atm*$= 1.13 \times 10^5$ Pa*?*

By the definition of bulk modulus,

$$\delta p = 2.2 \times 10^9 \,\mathrm{Pa} \times 0.01 = 2.2 \times 10^7 \,\mathrm{Pa}$$

which is 195 atm!

Example 3.3: *What is the speed of sound in water, if the density is* $1.00 \times 10^3 \,\mathrm{kg\,m^{-3}}$*?*

With the given value of B, it is $1.48 \,\mathrm{km\,s^{-1}}$.

3.3 *The speed of sound in an ideal gas*

Isothermal speed

If a layer in the gas column contracts/expands as the wave passes by, then its temperature increases/decreases relative to the adjacent layers. If this process is slow enough, there is time for heat to flow between adjacent layers, so that their temperature becomes the same. In this case, the temperature remains constant as the wave passes through the gas column, and the process is said to be isothermal. Then the product pV remains constant, according to the ideal gas law. This means that if for a layer of the gas column the pressure changes by δp and the volume changes by δV, as the wave passes through the layer,

$$V\delta p + p\delta V = 0 \Rightarrow \delta p = -p\frac{\delta V}{V} \tag{3.15}$$

If we compare this to the definition of bulk modulus, we see that B_{iso} is simply the pressure p in this case. Thus the **isothermal speed of sound** is

$$v_{iso} = \sqrt{\frac{p}{\rho}} \tag{3.16}$$

But according to the ideal gas law,

$$p = \rho\frac{kT}{m} \tag{3.17}$$

where k is Boltzmann's constant, T is the temperature (in Kelvin), and m is the mean molecular mass of the gas. So finally the isothermal speed of sound is

$$v_{iso} = \sqrt{\frac{kT}{m}} \tag{3.18}$$

Adiabatic speed of sound

If the process is fast enough that there is no time for heat to flow between adjacent layers, then the process follows the adiabatic law

$$pV^\gamma = \text{ constant} \tag{3.19}$$

where $\gamma = C_p/C_V$ is the **specific heat ratio**, which at normal temperatures has a value of $7/5$ for diatomic gases, such as the components of the Earth's atmosphere, and $5/3$ for monoatomic gases, such as the noble gases Ar or Ne.

In this case, a variation in the pressure of a layer δp and its associated change in volume δV, satisfy the condition

$$\frac{\delta p}{p} + \gamma \frac{\delta V}{V} \tag{3.20}$$

So in this case the bulk modulus is

$$B_{ad} = -\gamma p \tag{3.21}$$

and the **adiabatic speed of sound** is

$$v_{ad} = \sqrt{\frac{\gamma p}{\rho}} = \sqrt{\frac{\gamma kT}{m}} \tag{3.22}$$

In practice, the expansion and contraction of each layer is usually fast enough for the process to be adiabatic, and not isothermal.

Example 3.4: *Calculate the adiabatic speed of sound in air, composed of diatomic molecules of average molecular mass 30 u, at a temperature of* 0 °C.

For diatomic molecules $\gamma = 7/5$. Their mass in kilograms is 4.98 kg. We mustn't forget to convert the temperature to Kelvin, so $T = 273\,\text{K}$; and Boltzmann's constant is $k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$. Putting everything together,

$$v = 325\,\text{m s}^{-1}$$
